[2025-2] 공간정보분석 3: 시공간 데이터사이언스

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1 Introduction

1.1 Data Science

- definition
  - science of data: acquisition, management, handling, analysis of (un)structured data
  - science from data: extracting knowledge and insights from data
- data science is an iterative process as we refine questions based on data exploration and initial findings
- workflow
  - often preprocessing are required
  - Pareto principle: 80% of the results come from 20% of the efforts
  - data-information-knowledge pyramid
    - data $\longrightarrow$ information: organized into a databse, processed and analyzed
    - information $\longrightarrow$ knowledge: contextualized, understood, guided by questions and expertise
- uncertainty
  - measurement error
  - sampling error
  - processing error
  - model uncertatinty
- hierarchical modeling
  - phenomena are modeled at different levels of abstraction
  - three hierarchies to design a model with uncertainties
    - data model: expressing the distribution of the data
    - process model: modeling scientific uncertainty in the true process
    - parameter model: governing parameters of the model
  - scientific relationships in the process/parameter models can compensate the paucity of data

1.2 Spatiotemporal Data Science

- data science for and about data with space and time dimensions
  - extracting spatiotemporal features, developing spatiotemporal models
  - spatiotemporal dimensions are often implicit $\longrightarrow$ need to be extracted
- spatiotemporal data
  - spatial relationships + dynamics
  - violation of i.i.d. assumption

- two perspectives on dynamic spatial variability
  - object view
    - individual objects interact with each other
    - e.g. tracking moving vehicles, migration patterns, human moblity
  - field view
    - multivariate datum at every location
    - e.g. temprature maps, population density surfaces, land use classifications

1.3 Spatial Analysis Recap

- key concepts:
  - distance
  - connectivity
  - spatial weights
  - distance decay
  - autocorrelation

1.4 Temporal Data: The Fourth Dimension

- key elements of temporal data
  - granularity: second, minuite, hour, ...
  - extent
  - resolution: frequency of observations
  - topology: before, after, during, overlaps
  - serial autocorrelation
- quantifying time: POSIX representation
  - so called UNIX time
  - starts at 1970-01-01 00:00:00 UTC
  - counts time in integer after standard time
  - why?
    - universal reference $\longrightarrow$ consistency across systems
    - simplicity, precision, efficiency
    - timezone agnostic

2 Space-Time Data Analysis

2.1 Spatiotemporal Data Organization

- data structures
  - space-time cube: 3D representation (x, y, t)
  - trajectory data: moving objects
  - raster time series: stacks of rasters over time
  - event data: $(x, y, t, \text{attributes})$
  - panel data: repeated observations at fixed locations
- unit exchange: common framework to measure the variability over space and time

- normalization
- velocity-based
- variogram-based
- domain-specific

2.2 Spatiotemporal Data Representation

- Raster data
  - e.g. satelite imagery time series, climate model
  - typical formats: NetCDF, HDF5, GeoTIFF, GRIB2
- Vector data
  - points: trajectories
  - polygons: evolving boundaries
  - networks: dynamic networks
  - vector cubes

2.3 Major Challenges in Spatiotemporal Analysis

1. scale mismatch
   - spatial scale
   - temporal scale
   - cross-scale interactions: micro affects macro
2. computational complexity
   - curse of dimensionality
   - need of efficient data structures
3. statistical challenges
   - non-stationarity: properties change over space and time
   - anisotropy: directional dependencies (비등방성)
   - edge effects: boundary problems $\longrightarrow$ padding, windowing

2.4 Visualization

- visulization strategies
  - animation: time as frames
  - small multiples: facted by time (each panel = time)
  - space-time paths: 3D trajectories
  - Hovmöller diagram
  - isosurfaces: 3D version of contours
- slicing and indexing
  - types of slices
    - spatial slice (at fixed time) i.e. snapshot
    - temporal slice (at fixed location) i.e. time series
    - diagonal slice: space-time trajectory
    - aggregated slice: temporal/spatial average e.g. space-time windows

2.5 Spatiotemporal Centrography

- spatiotemporal extensions
  - mean center trajectory
  - space-time centroid
  - temporal weighted centers
  - space-time ellipsoid

2.6 Spatiotemporal Statistics

- spatial time series ⟶ physics-inspired or hierarchical dymanic modelling
  - spatiotemporal stationarity
  - spatiotemporal covariance
  - spatiotemporal variograms
  - spatiotemporal kriging
  - temporal autocorelation, tele-coupling (long-range correlation)
- spatiotemporal point process
  - extension of spatial point process
  - models: Poisson, Cox, cluster processes
  - tests: spatiotemporal K-function, scan statistics
- time series of lattice(areal) data
  - spatiotemporal autoregressive regression (STAR)
    - extends traditional autoregressive models to incorporate spatial/temporal dependencies
    - models the relationship between a variable and its past values + the values of neighboring locations

$$Y_{i,j,t} = \sum_{p=1}^{P}\sum_{q=1}^{Q} \phi_{p,q}\, Y_{i-p,\, j-q,\, t-1} + \sum_{r=1}^{R} \theta_r\, X_{i,j,t-r} + \epsilon_{i,j,t}.$$

    - typically, Q, R are 1 or 2 ⟶ STAR(1,1)
- outlier detection
  - spatiotemporal outlier: spatial outlier with spatiotemporal neighborhood
    - spatiotemporal neighborhoods
    - spatial time series
    - flow anomalies
      - dominant time intervals with high mismatch fraction
      - SWEET (smart window enumeration and evaluation of persistent thresholds)
- pattern
  - spatiotemporal association: instances close in both space and time
    - temporal ordering: co-occurrence < cascade < sequential
  - spatial colocation
    - statistical: cross-K with Monte-Carlo, mean NN distance, regression
    - data-mining: apriori-style transaction-based, distance-based
    - others: extended objects, rare events, regional colocation, fast algorithms
  - spatiotemporal event associations

  - co-occurrence: monotonic composite interest w/ filter-and-refine
  - cascades: metric & pruning
  - sequential patterns: sequence index
- trajectory-based associations
  - challenges: duration, direction, imprecision
  - colocation episodes: frequent sequences of colocation patterns
  - sequential patterns: segment-level region sequences
  - travel companions: groups of objects that move together
- spatial time series oscillation and tele-connections
  - finding pairs of spatial time series with correlation above a threshold
  - techniques: cone-tree index + filter-and-refine
  - testing: dipoles via wild bootstrap

2.7 Spatial Classification/Regression

1. spatial autocorrelation/dependency
   - feature creation
     - generate spatially contextual features
     - e.g. aggregation over neighborhoods, spatial association rules, spatial transformation of raw features
   - spatial interpolation
     - geostatistical: kriging (ordinary, simple, cokriging, universal)
     - non-geostatistical: nearest neighborhoods, IDW
     - hybrid methods
   - Markov random field (MRF)
     - labels depend only on neighboring labels
     - often combined with maximum likelihood clasifier
   - spatial autoregressive models
     - spatially autoregressive (SAR): $y = \rho W y + X\beta + \epsilon$
     - conditional autoregressive (CAR): equivalent to MRF (used in Bayesian hierarchical models)
   - spatial accuracy
     - traditional 0/1 loss $\longrightarrow$ ignores distance
     - penalizes misclassifications proportionally to distance
2. spatial heterogeniety
   - local/regional models
     - partition space into homogeneous zones, learn a separate model
     - learn a model at every location with spatial smoothing on parameters
   - GWR
     - coefficients become spatial functions $\beta(s)$
     - multi-model regression $\longrightarrow$ regularises neighboring parameters for smoothness
3. multi-scale effect (MAUP)
   - learn models at multiple candidate scales and compare
4. spatiotemporal autocorrelation

- STAR
- spatiotemporal kriging: covariance matrix & variograms across space-time
- relational probability trees / forests: tree node tests on spatial features and temporal changes
- Bayesian hierarchical models: explicitly encode dependence in the process layer

5. temporal non-stationarity
   - DSTM (dynamic spatiotemporal models): bayesian hierarchical / three-structure models
     - data model: observations and hidden processes
     - process model: spatial/temporal dependence
     - parameter model: priors
   - inference with Kalman filter (linear/Gaussian) or MCMC
6. prediction for moving objects
   - trajectory classification $\longrightarrow$ frequent sequential patterns within trajectories
   - location prediction: match current visit sequence ot historical sequences
   - location recommendation: rank candidate spots by popularity, user interests, collaborative patterns

2.8 Spatiotemporal Partitioning & Summarization

- partitioning
  - spatial: clustering of vector or raster items
  - spatiotemporal: grouping in 3-D(space-time)
  - after partitioning, we often summarize each cluster
  - challenges
    - noise and outliers
    - complex relationships
    - quality vs. efficiency trade-off
- spatial partitioning: by data types
  - point clustering
    - global: K-means, K-medoids, EM, CLIQUE, BIRCH, CLARANS
    - hierarchical: agglomerative / divisive, chameleon $\longrightarrow$ robust to outliers
    - density-based: DB-SCAN, ST-DBSCAN, Voronoi diagrams $\longrightarrow$ arbitrary shapes
  - polygon clustering
    - distance metrics: Hausdorff, overlap ratio, topology, attribute similarity
    - algorithms: K-means, CLARANS, Shared Nearest Neighbor
  - image segmentation (areal data)
    - guided by non-spatial attributes (color, texture)
    - methods: region growing, spllit-and-merge
  - network partitioning methods
    - network Voronoi diagram: partitions by closest node
    - capacity-constrained CCNVD: adds capacity while preserving contiuity
    - METIS: scalable graph partitioning
- spatiotemporal partitioning
  - 2D spatiotemporal events: ST-DBSCAN

- 3D grids: ST-GRID
- spatial time-series: K-means, K-medioids, EM+hierarchical
- trajectories: grouping, flock detection, segmentation

- spatiotemporal summarization
  - classical: aggregations
  - spatial: centroids, medioids $\longrightarrow$ K-means, K-medioids
  - network activities: primary routes $\longrightarrow$ K-Main Routes (KMR)
  - spatial time-series: remove autocorrelation $\longrightarrow$ traffic-stream summarizers, K-means centroids
  - trajectories: corridor extraction $\longrightarrow$ K-primary corridors

2.9 Spatiotemporal Hotspot Detection

- hotspot detection
  - find regions with unexpectedly high event density
  - foundation: scan statistics (null as homogeneous Poisson process)
  - challenges:
    - unknown location, size, shape, number of locations
    - avoid false positives
- point pattern
  - clustering: candidate hotspots from clusters $\longrightarrow$ test significance
  - scan statistics: varying window shapes and sizes
  - kernel density estimation: high density cells from density map
- spatial scan statistics

$$\lambda = \max_Z \left[\left(\frac{n_Z}{E[n_Z]} \right)^{n_Z} \left(\frac{N - n_Z}{N - E[n_Z]} \right)^{N - n_Z} \right]$$

  - Z: window
  - n_Z: events overlapped with the window
- spatiotemporal hotspots
  - network trajectories: hot routes
  - spatiotemporal scan: persistent vs. emerging

2.10 Spatiotemporal Change and Change Footprints

- taxonomy of change footprints
  - temporal: snapshot, set of snapshots, single point in series/interval
  - spatial: raster vs. vector
- raster-based patterns
  - local: pixel-wise $\longrightarrow$ pixel differencing
  - focal: depends on neighborhood $\longrightarrow$ block-based density
  - zonal: aggregated region $\longrightarrow$ object-based segmentation

2.11 Graphs and Networks

- graphs as natural language of spatial-network data

- topology dominates geometry
  - distance is defined by the network topology
  - violation of isotropy/symmetry
- high-dimensional, dynamic edge attributes
  - e.g. censors $\longrightarrow$ hundreds of attributes evolve over time
  - attributes attached to edges and nodes
- need for reference frame
  - moving agents experience the network in a Lagrangian view
  - concept of spatiotemporal graphs

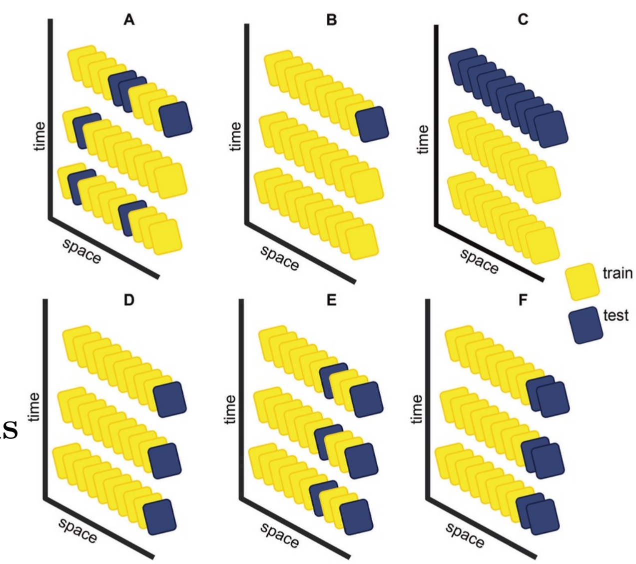
- network distance

$$d(i,j) = \min_{p \in P_{i,j}} \sum_{e \in p} w(e)$$

- network statistics, graph ploatforms

2.12 Performance Evaluation for Spatiotemporal Data

- generalizability
  - ultimate quality of a model
  - evaluation techinques, regularization
  - bias-variance trade-off: training error vs. validation error
    - bias: more flexible models will yield lower bias
    - variance: variability of model performance from many models
- spatial cross-validation
  - divide the dataset into K spatial folds
  - fold composition can be done by clustering, grid-based partitioning, or with buffers
- spatiotemporal cross-validation
  - temporlal dimension adds another layer
  - examples
    - randomly partitioned
    - leave-one-out
    - leave-time-out
    - leave-location-out
    - random k-fold leave-location-out
    - block leave-location-out



3 Statistical Times Series Analysis

3.1 Temporal Indexing and Definitions

- definitions
  - time series $\{X_t : t \in T\}$: a sequence of observations indexed by time
  - strict stationarity: for all h and k, $(X_{t_1}, \ldots, X_{t_k}) \stackrel{d}{=} (X_{t_1+h}, \ldots, X_{t_k+h})$
  - weak stationarity: $E(X_t) =$ constant and $\mathrm{Cov}(X_t, X_{t-h}) = \gamma(h)$ depends only on the lag h
- sampling frequency (hourly, monthly, yearly, ...) $\longrightarrow$ determines the index set

3.2 Autocorrelation and Partial Autocorrelation

- autocorrelation function (ACF)

$$\rho(h) = \gamma(h)/\gamma(0)$$

  - how correlated is the series with its post values
  - for stationary series, ACF decays as lags increase
  - for an MA(q) process, the ACF cuts off after lag q

- partial autocorrelation fuction (PACF)

$$\phi(h) = \mathrm{corr}(X_t, X_{t-h} \mid X_{t-1}, \ldots, X_{t-h+1})$$

  - how correlated the series with its past values after accounting for the values in between?
  - for an AR(p) process, the PACF cuts off after lag p

3.3 Explanatory Analysis

- visualization
  - trend: long term increase or decrease
  - seasonality: periodic fluctuations within a fixed period
  - outliers, structural breaks
  - periodicity
- seasonal-trend decomposition using LOESS (STL)
  - expresses a series as $X_t = T_t + S_t + R_t$

3.4 AR(I)MA Modelling

- AR(p)
  - self-dependent process of order p

$$X_t = \phi_1 X_t - 1 + \cdots + \phi_p X_{t-p} + \epsilon_t$$

- MA(q)
  - linear combination of past white noise terms of order q

$$X_t = \epsilon_t + \theta_1 \epsilon_t + \cdots + \theta_q \epsilon_{t-q}$$

- ARMA(p, q)
  - combined

$$X_t = \phi_1 X_t - 1 + \cdots + \phi_p X_{t-p} + \epsilon_t + \theta_1 \epsilon_t + \cdots + \theta_q \epsilon_{t-q}$$

- ARIMA(p, d, q)
  - includes differencing of order d
  - seasonal ARIMA: adds seasonal differencing and seasonal ARMA terms

3.5 Diagnostics

- unit-root

- if characteristic polynomial of an AR process has a root on the unit cycle, such series are non-stationary and require differencing
- e.g. AR(p)'s characteristic polynomial $\phi(x) = 1 - \phi_1 z - \cdots - \phi_p z^p$

- Dickey-Fuller test
  - rewrite AR(1): $\Delta X_t = X_t - X_{t-1} = \delta X_{t-1} + \epsilon_t$
  - null hypothesis: $\delta = 0$ (i.e. $\phi_1 = 1$)
  - test statistic: non-standard limit distribution, derived from functional CLT and Browninan motion
  - augmented DF test: adds lagged differences to remove serial correlation

$$\Delta X_t = \alpha + \beta t + \gamma X_{t-1} + \sum_{i=1}^{p-1} \delta_i \, \Delta X_{t-i} + \epsilon_t.$$

- Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test
  - reverse the hypothesis: null is that the series is level/trend-stationary
  - test statistic: partial sums of regression residuals and has a non-standard distribution involving Brownian bridges
- Phillips-Perron (PP) test
  - modify DF by allowing general forms of serial correlation and heteroscedasticity
  - same non-standard distribution as the DF test but uses kernel-based corrections

4 Advanced Time Series

4.1 Exponential Smoothing Methods

- exponential smoothing family
  - forecasting methods that apply exponentially decreasing weights to historical observations
  - equivalent to weighted MA w/ exponentially decaying weights
- simple exponential smoothing

$$\begin{aligned} \widehat{y}_{t+1|t} &= \alpha y_t + (1-\alpha)\widehat{y}_{t|t-1} \\ &= \alpha y_t + \alpha(1-\alpha) y_{t-1} + \alpha(1-\alpha)^2 y_{t-2} + \cdots \end{aligned}$$

  - $0 < \alpha < 1$: smoothing parameter (higher α gives more weight to recent observations)
  - the weights $\alpha(1-\alpha)^j$ decreases exponentially
- Holt's linear trend method
  - extends the simple exponential smoothing to handle trend (b_t)

$$\begin{aligned} \ell_t &= \alpha y_t + (1-\alpha)(\ell_{t-1} + b_{t-1}) \\ b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*) b_{t-1} \\ \widehat{y}_{t+h|t} &= \ell_t + h \cdot b_t \end{aligned}$$

    – l_t: level (절편)
    – b_t: trend (기울기)
    – α, β: level/trend smoothing parameter
    – h: forecast horizon
- Holt-Winters seasonal method

- adds seasonality to handle periodic patterns
- additive version:

$$\begin{aligned}
\ell_t &= \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}) \\
b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)b_{t-1} \\
s_t &= \gamma(y_t - (\ell_{t-1} + b_{t-1})) + (1-\gamma)s_{t-m} \\
\widehat{y}_{t+h|t} &= (\ell_t + h \cdot b_t) + s_{t+h-m}
\end{aligned}$$

  - m: seasonal period
  - γ: seasonal smoothing parameter
- multiplicative version:

$$\begin{aligned}
\ell_t &= \alpha \frac{y_t}{s_{t-m}} + (1-\alpha)(\ell_{t-1} + b_{t-1}) \\
b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)b_{t-1} \\
s_t &= \gamma \frac{y_t}{\ell_{t-1} + b_{t-1}} + (1-\gamma)s_{t-m} \\
\widehat{y}_{t+h|t} &= (\ell_t + h \cdot b_t) \times s_{t+h-m}
\end{aligned}$$

4.2 State Space Models

- linear Gaussian state space framework: unified framework for time series analysis with two fundamental equations
  - measurement (observation) equation:

$$y_t = Z_t \alpha_t + \epsilon_t$$

  - state (transition) equation:

$$\alpha_t = T_t \alpha_{t-1} + R_t \eta_t$$

    - y_t: $p \times 1$ observation vector
    - α_t: $m \times 1$ unobserved state vector
    - Z_t: $p \times m$ measurement matrix
    - T_t: $m \times m$ transition matrix
    - R_t: $m \times r$ selection matrix
    - ϵ_t, η_t: $p \times 1, r \times 1$ observation/state noise
- advantages of state space framework
  - unified framework: encompasses ARIMA, exponential smoothing, structural models
  - missing data: handling of irregular observations
  - multivariate: multiple series with shared components
  - filtering: real-time state estimation via Kalman filter
- examples
  - local linear trend model

$$\begin{aligned}
y_t &= \ell_t + \varepsilon_t, && \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2), \\
\ell_t &= \ell_{t-1} + b_{t-1} + \eta_t, && \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \\
b_t &= b_{t-1} + \zeta_t, && \zeta_t \sim \mathcal{N}(0, \sigma_\zeta^2).
\end{aligned}$$

$$\begin{aligned}
&(\text{관측식}) \quad y_t = \mathbf{Z}\,\alpha_t + \varepsilon_t, && \varepsilon_t \sim \mathcal{N}(0,\ \mathbf{H}), \\
&(\text{상태식}) \quad \alpha_t = \mathbf{T}\,\alpha_{t-1} + \mathbf{R}\,\eta_t, && \eta_t \sim \mathcal{N}(\mathbf{0},\ \mathbf{Q}).
\end{aligned}$$

$$\alpha_t = \begin{pmatrix} l_t \\ b_t \end{pmatrix}, \qquad \mathbf{Z} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad \mathbf{T} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{R} = \mathbf{I}_2,$$

- ARMA(1, 1): $y_t = \phi y_{t-1} + \theta \epsilon_{t-1} + \epsilon_t$

$$\begin{aligned} y_t &= s_t^{(1)}, \\ s_t^{(1)} &= \phi\, s_{t-1}^{(1)} + \theta\, s_{t-1}^{(2)} + \nu_t, \qquad \nu_t \sim \mathcal{N}(0, \sigma_\nu^2), \\ s_t^{(2)} &= \nu_t, \end{aligned}$$

$$\begin{aligned} (\text{관측식}) \quad y_t &= \mathbf{Z}\,\alpha_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \mathbf{H}), \\ (\text{상태식}) \quad \alpha_t &= \mathbf{T}\,\alpha_{t-1} + \mathbf{R}\,\eta_t, \qquad \eta_t \sim \mathcal{N}(0, \mathbf{Q}), \end{aligned}$$

$$\alpha_t = \begin{pmatrix} s_t^{(1)} \\ s_t^{(2)} \end{pmatrix}, \quad \mathbf{Z} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} \phi & \theta \\ 0 & 0 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

4.3 Kalmal Filter: Optimal State Estimation

- Kalman filter algorithm
  - Kalman filter provides the optimal linear estimator for the state vector in a linear Gaussian state space model
  - Kalman filter recursively computes the filtered state estimate $\hat{\alpha}_{t|t}$ and its uncertainty with covariance $P_{t|t}$
- prediction step (time update)
  - given information up to $t-1$, predict state and observation at time t:

$$\begin{aligned} \text{State prediction:} \quad & \widehat{\alpha}_{t|t-1} = T_t \widehat{\alpha}_{t-1|t-1}, \\ \text{State covariance:} \quad & P_{t|t-1} = T_t P_{t-1|t-1} T_t^\top + R_t Q_t R_t^\top, \\ \text{Observation prediction:} \quad & \widehat{y}_{t|t-1} = Z_t \widehat{\alpha}_{t|t-1}, \\ \text{Prediction error covariance:} \quad & F_t = Z_t P_{t|t-1} Z_t^\top + H_t. \end{aligned}$$

- update step (measurement update)
  - when observation y_t becomes available, update the state estimate:

$$\begin{aligned} \text{Prediction error:} \quad & \nu_t = y_t - \widehat{y}_{t|t-1}, \\ \text{Kalman gain:} \quad & K_t = P_{t|t-1} Z_t^\top F_t^{-1}, \\ \text{Updated state:} \quad & \widehat{\alpha}_{t|t} = \widehat{\alpha}_{t|t-1} + K_t \nu_t, \\ \text{Updated covariance:} \quad & P_{t|t} = P_{t|t-1} - K_t Z_t P_{t|t-1}. \end{aligned}$$

- properties of Kalman filter
  - optimality: minimizes MSE under Gaussian assumptions
  - recursiveness: only needs previous estimates and current observation
  - real-time: suitable for online filtering and forecasting
  - unbiasedness: $E[\alpha_{t|t}] = \alpha_t$ if model is correct
- log likelihood for parameter estimation: $\log L = -\frac{np}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^{n} \left[\log |F_t| + \nu_t^\top F_t^{-1} \nu_t \right].$

4.4 Advanced State Space Models

- unobserved components model
  - structural time series model: decomposes the series into interpretable components:

$$y_t = \mu_t + \gamma_t + \epsilon_t$$

  - each component can be modeled as a stochastic process using state space methods
- exponential smoothing methods are special cases of state space models
  - this connection allows:
    - automatic parameter selection via MLE
    - confidence intervals for forecasts
    - model comparison using information criteria
- Kalman smoothing and state estimation
  - forward-backward algorithm: while Kalman filter provides filtered estimates $\hat{\alpha}_{t|t}$, we often want smoothed estimates $\hat{\alpha}_{n|t}$ that uses all available data
- dynamic time warping (DTW)
  - measures similarity between sequences with different temporal alignments
  - algorithm: find optimal alignment path through cost matrix $C_{i,j} = d(x_i, y_j)$

$$DTW(X,Y) = \min_{\text{path}\pi} \sum_{(i,j)\in\pi} C_{i,j}$$

  - when comparing, standardizing each individually can stabilize the comparison
- link to deep learning approaches
  - key parallels
    - hidden states in RNNs $\leftrightarrow$ state vectors in state space models
    - memory mechanisms $\leftrightarrow$ Kalman filtering
    - sequential processing $\leftrightarrow$ recursive estimation
    - attention mechanisms $\leftrightarrow$ optimal weighting
  - recurrent neural networks (RNNs) vs. state space model

$$\begin{aligned} h_t &= \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b_h)\,, \\ y_t &= W_{hy}h_t + b_y. \end{aligned}$$

    - similarities: both maintain hidden state, sequential processing
    - differences: RNN uses nonlinear activation, learned parameters
- long short-term memory (LSTM)

$$\begin{aligned}
f_t &= \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) && \text{(forget gate)} \\
i_t &= \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right) && \text{(input gate)} \\
\tilde{C}_t &= \tanh\left(W_C \cdot [h_{t-1}, x_t] + b_C\right) && \text{(candidate)} \\
C_t &= f_t * C_{t-1} + i_t * \tilde{C}_t && \text{(cell state)} \\
o_t &= \sigma\left(W_o \cdot [h_{t-1}, x_t] + b_o\right) && \text{(output gate)} \\
h_t &= o_t * \tanh(C_t) && \text{(hidden state)}
\end{aligned}$$

- addresses vanishing gradient problem with gating mechanisms
- cell state C_t acts like a memory buffer, which passes long-term information
- hidden state h_t is the output at each time step
- both Kalman filter and LSTM selectively update and forget information

- gated recurrent unit (GRU)

$$
\begin{aligned}
r_t &= \sigma\left(W_r \cdot [h_{t-1}, x_t]\right) && \text{(reset gate)} \\
z_t &= \sigma\left(W_z \cdot [h_{t-1}, x_t]\right) && \text{(update gate)} \\
\tilde{h}_t &= \tanh\left(W \cdot [r_t * h_{t-1}, x_t]\right) && \text{(candidate)} \\
h_t &= (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t && \text{(hidden state)}
\end{aligned}
$$

- reset gate r_t controls how much past information to forget
- update gate z_t acts like Kalman gain: determines how much to update/retain

- deep learning methods (vs. classical methods)
  - nonlinear patterns can be captured
  - large datasets fully utilized
  - multivariate relationships learned automatically
  - complex temporal dependencies modeled
  - when performance is more important than interpretability

4.5 Multivariate Time Series

- cointegration
  - cointegration: statistical property of a collection of time series which, when individually non-stationary, can be combined to form a stationary series
  - if two or more time series are cointegrated, they share a common stochastic drift
- vector autoregression (VAR)
  - VAR model generates univariate autogregression to multivariate time series
  - each variable is a linear function of past values of itself and past values of all other variables

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + \epsilon_t$$

  - trend terms can be included as deterministic components
- distributed lag model: related to single equations from VAR: allow both short- and long-run dynamics
  - DL: regresses the dependent variable on current and past values of an explanatory variable

$$y_t = a + w_0 x_t + w_1 x_{t-1} + \cdots + w_L x_{t-L} + \varepsilon_t$$

  - ARDL: includes lags of the dependent variable

$$y_t = \mu + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=0}^{q} \theta_j x_{t-j} + \varepsilon_t$$

    – focus on exogenous variables and selecting the correct lag structure
- Granger causality: predictive notion of causality
  - a time series X Granger-causes Y if past values of X contain statistically significant information

about future values of Y beyond the information contained in past values of Y alone

- test: involves estimating VAR models with and without lagged X terms and performing an F-test on the lagged coefficients
- if two variables are cointegrated, at least one must Granger-cause the other ($\because$ the error-correction term conveys long-run equilibrium adjustments)

4.6 Spatial Time Series

- STAR
  - incorporating spatial autocorrelation

$$y_t = \rho W y_t + \phi y_{t-1} + \epsilon_t$$

    - W: spatial weight matrix
    - ρ: measures spatial dependence
- STARIMA
  - include spatial and temporal lags

$$y_t \; = \; \sum_{p=1}^{P} \Phi_p \, y_{t-p} \; + \; \sum_{p=0}^{P} \sum_{k=1}^{K} \Theta_{p,k} \, W^k \, y_{t-p} \; + \; \varepsilon_t.$$

    - Φ: coefficients for the temporal lags
    - Θ: coefficients for the spatial lags

5 New Architectures for Space-Time Data

5.1 Space-Time Data Structures

- tabular format
  - long format: each row = observation
  - repeated spatial locations over time
  - multiple features per location-time
- tensor format
  - advantages
    - efficient GPU computation
    - natural deep learning input format
    - preserves spatiotemporal structure
- from table to tensors
  - mini-batch processing is typical in deep learning $\longrightarrow$ a batch dimension comes first
  - space(2D or 3D) and time(1D) dimensions can be switched
  - transformation pipeline:

$$\text{DataFrame}(N \times D) \rightarrow \text{Tensor}(B \times T \times S \times F)$$

- handling irregular space-time data
  - missing time points, variable spatial coverage, misalignment $\longrightarrow$ interpolation, masking, padding

5.2 Spatiotemporal Data Linkage

- challenges
  - multiple data sources
  - temporal/spatial misalignment
  - entity resolution: matching locations across datasets
- methods
  - distance-based matching
  - spatial interpolation

5.3 Deep Learning Architectures for Spacetime

- gated recurrent unit (GRU)
  - simplified LSTM

$$
\begin{aligned}
z_t &= \sigma\left(W_z x_t + U_z h_{t-1} + b_z\right) && \text{(update gate)} \\
r_t &= \sigma\left(W_r x_t + U_r h_{t-1} + b_r\right) && \text{(reset gate)} \\
\tilde{h}_t &= \tanh\left(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h\right) && \text{(candidate)} \\
h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t && \text{(hidden state)}
\end{aligned}
$$

    – U_z, U_r, U_h: recurrent weight matrices
    – W_z, W_r, W_h: input weight matrices
    – b_z, b_r, b_h: biases
    – h_t: hidden state at time t
- TabPFN: for tabular space-time data
  - tabular prior-data fitted networks
  - pre-trained transformer that learns to do tabular prediction in a single forward pass
  - architecture:
    – transformer-based prior over tabular functions
    – no backpropagation needed for new datasets
    – fast inference on small-medium datasets
- connection to Gaussian processes
  - Gaussian process defines a distribution over functions:

$$
f(s,t) \sim \mathcal{GP}\left(m(s,t),\ k\left((s_i,t_i),(s_j,t_j)\right)\right)
$$

  - separable spatiotemporal covariance: $k\left((s_i,t_i),(s_j,t_j)\right) = k(s_i,s_j) \cdot k(t_i,t_j)$
  - common choices of kernel functions
    – spatial: Matérn, exponential
    – temporal: AR, seasonal
  - prediction(kriging): $y(\hat{s_o}) = k_o^\top K^{-1} y$
- graph neural networks (GNNs)
  - GNNs encode spatial relationships

$$
\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})
$$

    – nodes $\mathcal{V}$: spatial locations

  - edges $\mathcal{E}$: spatial proximity/relationships
  - adjacency matrix $A_{ij} = w(s_i, s_j)$
- message passing:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \frac{1}{c_{ij}}\mathbf{W}^{(l)}\mathbf{h}_j^{(l)}\right)$$

  - c_{ij} is a normalization constant, σ is an activation constant
  - $\mathcal{N}(i)$: neighbors of node i
- spatiotemporal GNN

$$\mathbf{H}_{t+1} = \mathrm{GRU}\big(\mathbf{H}_t,\ \mathrm{GCN}(\mathbf{A}, \mathbf{X}_t)\big)$$

  - $\mathbf{H}$: hidden state
  - $\mathbf{X}$: input features at time t

- SSM, GP, GNN: how to encode dependencies?

	State Space	**Gaussian Process**	**Graph Neural Network**
Representation	$h' = Ah + Bx$	$f \sim \mathcal{GP}(m, k)$	$h_i = \sigma\left(\sum_j A_{ij}h_j\right)$
Complexity	$\mathcal{O}(n)$	$\mathcal{O}(n^3)$ (커널 역행렬 계산)	$\mathcal{O}(\lVert\mathcal{E}\rVert)$ (edge 수에 비례)
Strength	Sequential	Uncertainty	Relational
Spatiotemporal	Markov chain	Covariance (공분산 구조)	Message passing

6 Causal Inference

6.1 Treatment Effects

- fundamental problem of causal inference
  - we observe either outcome under treatment or control, but never both in observational data
- treatment effects
  - ITE: in randomized controlled trial (RCT)

$$\tau_i = Y_i(1) - Y_i(0)$$

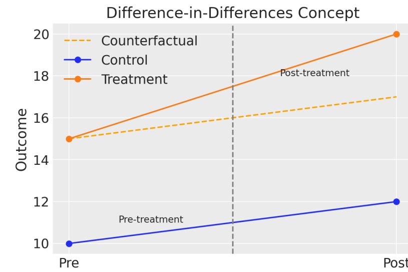
  - ATE: in observational studies

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]$$

    - key assumption: unconfoundedness (treatment assignment T is independent of potential outcomes given covariates X)

$$Y(1), Y(0) \perp\!\!\!\perp T \mid X$$

- defining treatments in spatiotemporal data
  - treatment: a condition or intervention applied to units over space and time
  - e.g. policy changes, environmental exposures, etc.
  - spatiotemporal consideration:
    - spatial spillover
    - temporal dynamics: treatment effects evolve over time



  - interference: SUTVA violation due to interactions or spillovers

6.2 DiD, PSM, SCM, SC, Causal ML

- difference-in-differences (DiD)
  - Classic 2×2 DiD setup:

$$Y_{it} = \alpha + \beta \cdot \text{Treated}_i + \gamma \cdot \text{Post}_t + \delta \cdot (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

  - Parallel trends assumption:

$$\mathbb{E}[Y_{i,t}(0) - Y_{i,t-1}(0) \mid \text{Treat} = 1] = \mathbb{E}[Y_{i,t}(0) - Y_{i,t-1}(0) \mid \text{Treat} = 0]$$

  - Estimator:

$$\hat{\delta} = (\bar{Y}_{\text{treat, post}} - \bar{Y}_{\text{treat, pre}}) - (\bar{Y}_{\text{control, post}} - \bar{Y}_{\text{control, pre}})$$

- DiD with staggered treatment
  - staggered adoption: different units treated at different times
  - problem with two-way-fixed-effects:

$$Y_{it} \;=\; \alpha_i + \lambda_t + \delta D_{it} + \varepsilon_{it}$$

    - negative weights problem
    - treatment effect heterogeneity bias
  - modern approach: event study

$$Y_{it} \;=\; \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbb{1}[\text{RelTime}_{it} = k] + \varepsilon_{it}$$

- propensity score matching (PSM)
  - balance covariates between treated and control groups
  - propensity score: $e(X_i) = P(T_i = 1 \mid X_i)$
  - Rosenbaum-Rubin theorem: if unconfoundedness holds given X, it also holds given $e(X)$
  - checking balance after matching
    - standardized mean difference (SMD)

$$\text{SMD} = \frac{\bar{X}_{\text{treat}} - \bar{X}_{\text{control}}}{\sqrt{(s^2_{\text{treat}} + s^2_{\text{control}})/2}}$$

    - systematic balancing: covariate balancing propensity score (CBPS)
- structural causal models (SCMs)
  - DAG(directed acyclic graph): represents causal structure
  - components:
    - endogenous variables
    - exogeneous variables
    - structural equations
- synthetic control method
  - for comparative case studies with few treated units

- find weights $\mathbf{w}$ such that:

$$\hat{Y}_{1t} = \sum_{j=2}^{J+1} w_j Y_{jt}, \qquad \sum_{j=2}^{J+1} w_j = 1,\ w_j \geq 0$$

- minimize pre-treatment difference:

$$\min_w \sum_{t=1}^{T_0} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt} \right)^2$$

- causal ML: double machine learning
  - for high-dimensional confounders

$$Y = \theta T + g(X) + \varepsilon_Y,$$
$$T = m(X) + \varepsilon_T,$$

  - causal ML algorithm:
    - split data into folds
    - fit $\hat{g}$ and $\hat{m}$ using ML on training fold
    - copute residuals on test fold: $\tilde{Y} = Y - \hat{g}(X),\ \tilde{T} = T - \hat{m}(X)$
    - estimate $\hat{\theta}$ from $\tilde{Y} = \theta\tilde{T} + \epsilon$

6.3 Challenges

- SUTVA
  - stable unit treatment value assumption
    - no interference between units
    - consistency of potential outcomes
- violation
  - spillover effects
    - solutions: spatial DiD with spillover buffers, network-based causal inference, moran's I test for spatial autocorrelation
  - time-varying confounding
    - inverse probability weighting, G-computation, marginal structure models

7 Text Data

7.1 Variational Autoencoder

- decomposition
  - autoencoder: a neural network that compresses input data and the reconstructs it
  - variational models: learning probabilistic latent representations
    - probabilistic: aim to learn idstributions for data generation
    - latent: hidden variables that capture underlying factors (i.e. topics, styles, ..)
    - representation: want to learn meaningful features (i.e. embeddings)
  - VAE is essentially based on variational inference (class of Bayesian methods)

- variational inference
  - objective: approximate posterior distribution $p(z \mid x)$ of data x
  - evidence lower bound (ELBO) $\longrightarrow$ maximize

$$\log p(x) \;\geq\; \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)]}_{\text{reconstruction term}} - \underbrace{D_{\mathrm{KL}}\big(q_\phi(z \mid x) \,\|\, p(z)\big)}_{\text{regularization (prior matching)}}$$

    – $q_\phi(z \mid x)$: approximate posterior to learn (encoder)
    – $p_\theta(x \mid z)$: generative distribution (decoder)
    – $p(z)$: prior distribution
- reparameterization
  - required to enable backpropagation
  - redefine the sampling operation to a generative and differentiable function
  - when encoder output is $\mu(x), \sigma(x)$, sampling: $z = \mu(x) + \sigma(x) \odot \epsilon$

7.2 BERT

- bidrectional encoder representations from trannsformers
  - pretrained language model that captures context in text

7.3 Visualization

- t-SNE: t-distributed stochastic neighbor embedding

7.4 Document Similarity Metrics

- cosine similarity

$$\text{similarity}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \, \|B\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}$$

- metric using euclidean distance

$$\text{similarity}(A, B) = \frac{1}{1 + d(A, B)} \quad \text{where } d(A, B) = \sqrt{\sum_{i=1}^{n} (A_i - B_i)^2}$$